DA-013 • ENVIRONMENTAL ANALYTICS

Air Quality & Pollution Pattern Analysis

Comparing pollutant distributions across dates and monitoring locations for urban planning

Project Scope & Target Users

CORE OBJECTIVE

Explore air quality measurements to compare pollutant levels across dates and stations.

STAKEHOLDER NEED

Students and local planners summarize how pollution varies by time and location.



Key Pollutants Measured

PM2.5 & PM10

Fine inhalable particulate concentrations.

NO_2 & SO_2

Traffic and industrial combustion gases.

Ground Ozone

Photochemical urban smog indicator.

Composite AQI

Standardized daily health risk score.

Analytical Methodology

01

Data Cleaning

Handle missing sensor readings and outliers.

02

Distribution Audit

Inspect pollutant spread across stations.

03

Temporal Grouping

Compare daily, monthly, and seasonal shifts.

04

Hotspot Detection

Identify peak pollution events and zones.

CORE ANALYTICAL FINDINGS

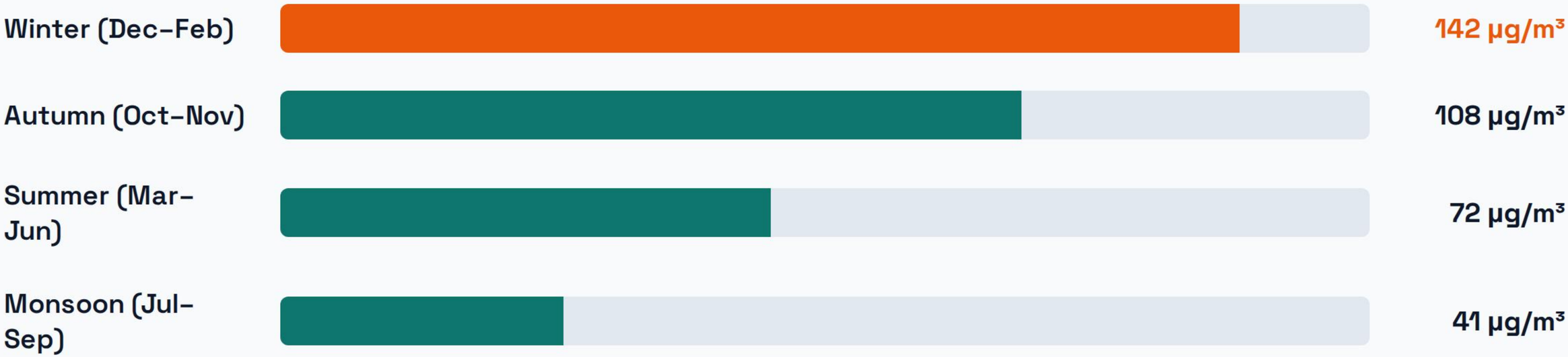
Spatial & Temporal Pollution Patterns

WHO Safe Limit Exceedance

3.4×

Peak winter PM2.5 concentration above annual safe thresholds

Seasonal PM2.5 Concentration Trends



Industrial vs. Residential Zone Variance

INDUSTRIAL & TRAFFIC HUBS

High Baseline: Sustained elevated PM10 and NO_2 levels.

Rush Hour Spikes: Morning and evening traffic surges.

Frequent Alerts: 42% of days exceed safe AQI.

SUBURBAN & GREEN ZONES

Lower Variance: Stable moderate daily readings.

Weather Driven: Wind speed disperses local particulates.

Clean Baseline: 81% of days within safe limits.

Primary Drivers of High-Pollution Records



Vehicular Exhaust: Primary driver of weekday NO_2 peaks.

Industrial Output: Continuous point-source particulate emissions.

Thermal Inversion: Cold air traps pollutants near ground.

Construction Dust: Localized coarse PM10 surges.

Dataset Summary Metrics

24

Active urban monitoring stations analyzed

68%

Higher AQI recorded during peak winter months

08:00

Morning rush-hour peak across city corridors

Recommendations for Local Planners

Low-Emission Zones: Restrict heavy diesel vehicles in hotspots.

Green Buffers: Plant dense tree belts near industrial corridors.

Early Warning Alerts: Issue health advisories before winter peaks.

Sensor Expansion: Deploy low-cost nodes in underserved wards.



Thank You & Questions

DA-013: Air Quality and Pollution Pattern Analysis • Environmental Analytics

Image Sources



https://www.undp.org/sites/g/files/zskgke326/files/styles/image_with_credit_caption_wide_large_1920_x897_/public/2025-02/undp-gctisd-airpollutionblog4pic1-2025.png?itok=xh4DPfB5

Source: www.undp.org



https://images.pexels.com/photos/36011163/pexels-photo-36011163.jpeg?cs=srgb&dl=pexels-wal_-172619-2156618639-36011163.jpg&fm=jpg

Source: www.pexels.com



https://unity.edu/wp-content/uploads/2025/03/Green-City-scaled-1-1024x576.jpg.webp

Source: unity.edu